

When to Use

To test if two or more populations or subpopulations have different patterns of response to a qualitative/categorical DV. This can also be characterized as a test of a pattern of relationship between two qualitative/categorical variables.

H Research Hypothesis: The researcher hypothesized that films and serials would tend to have larger age gaps between the actors playing Lois Lane and Clark Kent (Average or Big), whereas TV shows would tend to have minimal age gaps.

H Null Hypothesis: There is no pattern of relationship between media type (Film/Serial vs TV Show) and age gap category (Minimal vs Average/Big) in the population of Superman portrayals represented by this sample.

Running a Chi-Square Test

Analyze → Descriptive Statistics → Crosstabs

- Highlight the variable you want to define the rows (be sure it is qualitative/categorical) and click arrow
- Highlight the variable you want to define the columns (be sure it is qualitative/categorical) and click arrow
- “Statistics” — check that you want a “Chi-square analysis”

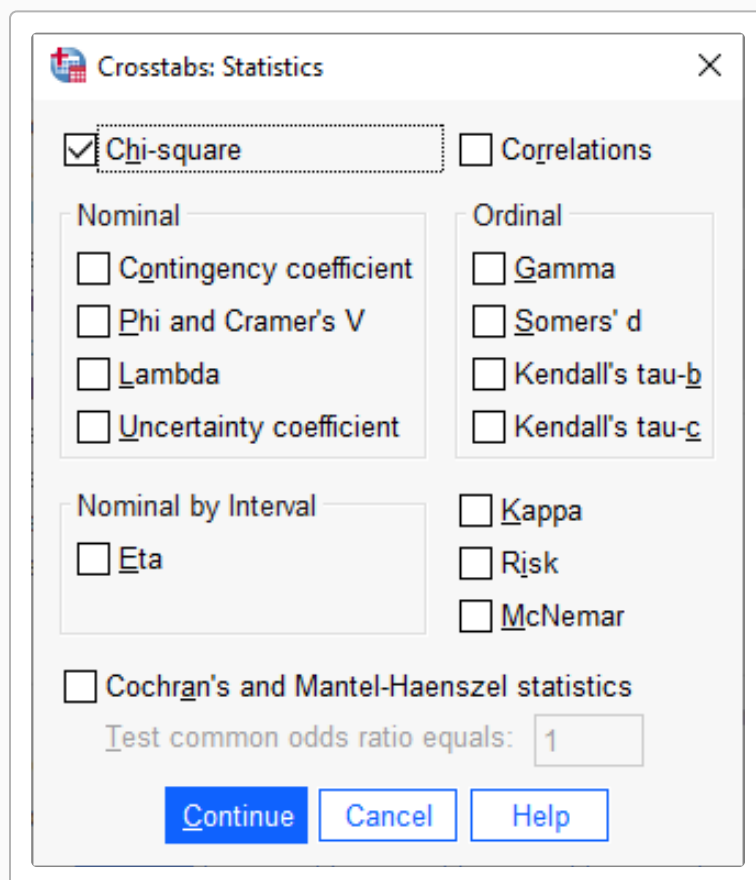
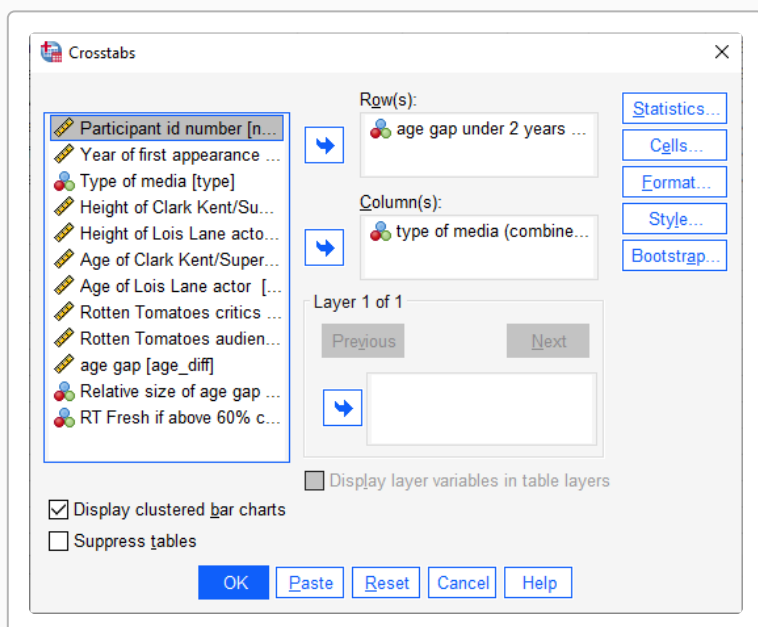
SPSS Syntax

CROSSTABS

```
/TABLES=type2 BY age_grp2
/STATISTICS=CHISQ
/CELLS=COUNT ROW COLUMN TOTAL.
```

①
②
③

- ① Use "row variable" BY "column variable"
- ② Get Chi-square significance test
- ③ Get various row, column and/or total cell percentages (optional)



SPSS Output

Chi-Square Tests			
	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	5.143	1	.023
Continuity Correction	2.009	1	.156
N of Valid Cases	9		

4 cells (100.0%) have expected count less than 5. The minimum expected count is 0.67.
Computed only for a 2x2 table.

type2 * age_grp2 Crosstabulation				
type2		Average/Big	Minimal	Total
Film/Serial	Count	6	0	6
	Expected Count	4.7	1.3	6.0
TV Show	Count	1	2	3
	Expected Count	2.3	0.7	3.0
Total	Count	7	2	9
	Expected Count	7.0	2.0	9.0

Reporting the Results

It is important to show the cell and marginal frequencies (often also percentages) in a contingency table before presenting the Chi-square results.

As in the example, be sure to communicate:

- The research hypothesis (if there is one)
- The statistical results
- Whether or not those results support the research hypothesis

The p-value of 0.023 means that there is about a 2.3% chance that this result is a Type I error.

Remember, even if the printout shows it, never report $p = .000$, because that would suggest there is no possibility of a Type I error. Instead, report " $p < .001$ ".

Chi-square results are "suspicious" if more than 15% of the cells have expected frequencies less than 5.

The "correction for continuity" is computed for 2x2 designs, to give a better estimate of Type I error for this small design, especially when N is small.

Table 1

Crosstabulation of Media Type and Age Gap Category

Media Type	Age Gap Category		
	Minimal	Average/Big	Total
Film/Serial	6 (100%)	0 (0%)	6 (100%)
TV Show	1 (33.3%)	2 (66.7%)	3 (100%)
Total	7	2	9

Note. $\chi^2(1) = 5.14$, $p = .023$.

Example Write-Up

Sample APA Write-Up

Table 1 shows the contingency table for these variables. The sample of Superman portrayals was divided between the two types of media (Film/Serial and TV Show) and between the two types of age gap categories (Minimal and Average/Big). Inspection of the table suggests that, as hypothesized, films and serials tended to have larger age gaps (Average or Big), whereas TV shows tended to have minimal age gaps. Consistent with the hypothesis, there was a statistically significant relationship between the variables, $\chi^2(1) = 2.01$, $p = 0.156$, when the correction for continuity was applied to this 2x2 design.

i Note

This example shows how to structure a write-up following the guidelines: stating the hypothesis, describing the table, reporting the chi-square results, and interpreting the findings in relation to the hypothesis.